

Esercitazione Comunicazione tra due macchine tramite server dns e host assegnato, e intercettazione comunicazione tramite WireShark

Comunicazione HTTPS dove si evidenziano IP e MAC Sorgente e Destinazione:

Con l'HTTPS, prima di trasferire i dati il browser e il server stabiliscono una connessione sicura e crittografata.

L'utente visita un sito Web HTTPS digitando l'URL in formato https:// nella barra degli indirizzi del browser.

Il browser prova a verificare l'autenticità del sito richiedendo il certificato SSL del server.

Il server Web invia il certificato SSL/TLS che contiene una chiave pubblica come risposta.

Il certificato SSL del sito Web dimostra l'identità del server. Una volta che il browser è soddisfatto del certificato, utilizza la chiave pubblica per crittografare e inviare un messaggio che contiene una chiave di sessione segreta.

Il server Web utilizza la propria chiave privata per decrittografare il messaggio e recuperare la chiave di sessione. Quindi utilizza la chiave di sessione per crittografare e inviare un messaggio di conferma al browser.

Dopo di ciò sia il Client(browser) che il Server utilizzano la stessa chiave (di sessione) per cifrare i dati scambiati.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	192.168.32.101	192.168.32.255	BROWSER	252	Domain/workgroup Announcement WORKGROUP, NT Workstation, Domain Enum
2	2.488887555	192.168.32.101	192.168.32.100	TCP	66	49162 → 443 [SYN] Seq=0 Win=0 Len=0 MSS=1460 SACK_PERM WS=128
3	2.488874647	192.168.32.100	192.168.32.101	TCP	66	443 → 49162 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM WS=128
4	2.489343567	192.168.32.101	192.168.32.100	TCP	60	49162 → 443 [ACK] Seq=1 Ack=1 Win=65700 Len=0
5	2.489921628	192.168.32.101	192.168.32.101	TLSv1	215	Client Hello (Client to Server)
6	2.489921628	192.168.32.100	192.168.32.101	TCP	54	443 → 49162 [ACK] Seq=1 Ack=162 Win=64128 Len=0
7	2.522803889	192.168.32.100	192.168.32.101	TLSv1	1373	Server Hello, Certificate, Server Key Exchange, Server Hello Done
8	2.532107153	192.168.32.101	192.168.32.100	TLSv1	188	Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message
9	2.532780995	192.168.32.100	192.168.32.101	TLSv1	113	Change Cipher Spec, Encrypted Handshake Message
10	2.551706335	PCSSystemtec_d6:02::	Broadcast	ARP	60	Who has 192.168.32.1? Tell 192.168.32.101
11	2.745325044	192.168.32.101	192.168.32.100	TCP	60	49162 → 443 [ACK] Seq=290 Ack=1379 Win=64320 Len=0
12	3.230631331	PCSSystemtec_d6:02::	Broadcast	ARP	60	Who has 192.168.32.1? Tell 192.168.32.101
13	4.228385937	PCSSystemtec_d6:02::	Broadcast	ARP	60	Who has 192.168.32.1? Tell 192.168.32.101
14	4.234969102	192.168.32.101	192.168.32.255	BROWSER	243	Local Master Announcement WINDOVS-7, Workstation, Server, NT Workstation, Potential Browser, Master Browser
15	5.698050457	fe80::984d:d486:dab...	ff02::1:3	LLNMR	84	Standard query 0x2514 A wpad
16	5.698050619	192.168.32.101	224.0.0.252	LLNMR	64	Standard query 0x2514 A wpad
17	5.790884671	fe80::984d:d486:dab...	ff02::1:3	LLNMR	84	Standard query 0x2514 A wpad
18	5.790884915	192.168.32.101	224.0.0.252	LLNMR	64	Standard query 0x2514 A wpad
19	5.995040971	192.168.32.101	192.168.32.255	NBNS	92	Name query NB WPA0<00>
20	6.744602970	192.168.32.101	192.168.32.255	NBNS	92	Name query NB WPA0<00>
21	7.493356861	192.168.32.101	192.168.32.255	NBNS	92	Name query NB WPA0<00>
22	7.537649080	PCSSystemtec_17:9c::	PCSSystemtec_d6:02::	ARP	42	Who has 192.168.32.101? Tell 192.168.32.100
23	7.538637464	PCSSystemtec_d6:02::	PCSSystemtec_17:9c::	ARP	60	192.168.32.101 is at 08:00:27:d6:02:dd

Frame 5: 215 bytes on wire (1720 bits), 215 bytes captured (1720 bits) on interface eth0, id 0

- Ethernet II, Src: PCSSystemtec_d6:02:dd (08:00:27:d6:02:dd), Dst: PCSSystemtec_17:9c:a3 (08:00:27:17:9c:a3)
 - Destination: PCSSystemtec_17:9c:a3 (08:00:27:17:9c:a3)
 - Source: PCSSystemtec_d6:02:dd (08:00:27:d6:02:dd)
 - Type: IPv4 (0x0800)
- Internet Protocol Version 4, Src: 192.168.32.101, Dst: 192.168.32.100
 - Transmission Control Protocol, Src Port: 49162, Dst Port: 443, Seq: 1, Len: 161
 - Transport Layer Security
 - TLSv1 Record Layer: Handshake Protocol: Client Hello
 - Content Type: Handshake (22)
 - Version: TLS 1.0 (0x0301)
 - Length: 156
 - Handshake Protocol: Client Hello

Comunicazione HTTP dove si evidenziano IP e MAC Sorgente e Destinazione:

I messaggi HTTP sono in chiaro, il che significa che le parti non autorizzate possono accedervi e leggerli facilmente su Internet

The screenshot shows a Wireshark capture of network traffic on the eth0 interface. The packet list at the top shows a series of packets, with packet 103 selected. The packet details pane on the left shows the structure of the selected packet, including Ethernet II, Internet Protocol Version 4, and Hypertext Transfer Protocol. The packet bytes pane on the right shows the raw data in hexadecimal and ASCII. The Hypertext Transfer Protocol section is expanded, showing the request line: GET / HTTP/1.1. The packet list shows the source and destination IP addresses and ports for each packet. The packet details pane shows the source and destination MAC addresses and the protocol used. The packet bytes pane shows the raw data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
97	311.596975237	192.168.32.101	192.168.32.100	TCP	66	49170 → 80 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=4 SACK_PERM
98	311.59697293	192.168.32.100	192.168.32.101	TCP	66	80 → 49170 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM WS=128
99	311.597624421	192.168.32.101	192.168.32.100	TCP	66	49170 → 80 [ACK] Seq=1 Ack=1 Win=65700 Len=0
100	311.598893703	192.168.32.101	192.168.32.100	HTTP	444	GET / HTTP/1.1
101	311.598984503	192.168.32.101	192.168.32.101	TCP	54	80 → 49170 [ACK] Seq=1 Ack=391 Win=64128 Len=0
102	311.607924974	192.168.32.100	192.168.32.101	TCP	204	80 → 49170 [PSH, ACK] Seq=1 Ack=391 Win=64128 Len=150 [TCP segment of a reassembled PDU]
103	311.609259100	192.168.32.100	192.168.32.101	HTTP	312	HTTP/1.1 200 OK (text/html)
104	311.609781590	192.168.32.101	192.168.32.100	TCP	66	49170 → 80 [ACK] Seq=391 Ack=410 Win=65292 Len=0
105	311.610440351	192.168.32.101	192.168.32.100	TCP	66	49170 → 80 [FIN, ACK] Seq=391 Ack=410 Win=65292 Len=0
106	311.610453605	192.168.32.100	192.168.32.101	TCP	54	80 → 49170 [ACK] Seq=410 Ack=392 Win=64128 Len=0
107	311.642753909	192.168.32.101	192.168.32.100	TCP	66	49171 → 80 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=4 SACK_PERM
108	311.642798332	192.168.32.100	192.168.32.101	TCP	66	80 → 49171 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM WS=128
109	311.643995300	192.168.32.101	192.168.32.100	HTTP	320	GET /favicon.ico HTTP/1.1
110	311.644027550	192.168.32.100	192.168.32.101	TCP	54	80 → 49171 [ACK] Seq=1 Ack=267 Win=64128 Len=0
111	311.655095585	192.168.32.100	192.168.32.101	TCP	207	80 → 49171 [PSH, ACK] Seq=1 Ack=267 Win=64128 Len=153 [TCP segment of a reassembled PDU]
112	311.656087462	192.168.32.100	192.168.32.101	HTTP	252	HTTP/1.1 200 OK (image/x-icon)
113	311.657228084	192.168.32.101	192.168.32.100	TCP	66	49171 → 80 [ACK] Seq=267 Ack=353 Win=65348 Len=0
114	311.657228213	192.168.32.101	192.168.32.100	TCP	66	49171 → 80 [FIN, ACK] Seq=267 Ack=353 Win=65348 Len=0
115	311.657256742	192.168.32.100	192.168.32.101	TCP	54	80 → 49171 [ACK] Seq=353 Ack=268 Win=64128 Len=0
116	316.831402451	PCSSystemtec.17:9c:a3	PCSSystemtec.d6:02:dd	ARP	42	Who has 192.168.32.101? Tell 192.168.32.100
117	316.832027916	PCSSystemtec.d6:02:dd	PCSSystemtec.17:9c:a3	ARP	60	192.168.32.101 is at 08:00:27:d6:02:dd
118	355.361952252	192.168.32.101	192.168.32.255	BROWSER	252	Domain/Workgroup Announcement WORKGROUP, NT Workstation, Domain Enum

Frame 103: 312 bytes on wire (2496 bits), 312 bytes captured (2496 bits) on interface eth0, id 0

- Ethernet II, Src: PCSSystemtec.17:9c:a3 (08:00:27:17:9c:a3), Dst: PCSSystemtec.d6:02:dd (08:00:27:d6:02:dd)
- Destination: PCSSystemtec.d6:02:dd (08:00:27:d6:02:dd)
- Source: PCSSystemtec.17:9c:a3 (08:00:27:17:9c:a3)
- Type: IPv4 (0x800)
- Internet Protocol Version 4, Src: 192.168.32.100, Dst: 192.168.32.101
- Transmission Control Protocol, Src Port: 80, Dst Port: 49170, Seq: 151, Ack: 391, Len: 258
- [2 Reassembled TCP Segments (408 bytes): #102(150), #103(258)]
- Hypertext Transfer Protocol
- Line-based text data: text/html (10 Lines)

Frame (312 bytes) Reassembled TCP (408 bytes)

Protocolli http e https utilizzano inoltre porte diverse (80 il primo, 443 il secondo)